

GRADING RUBRIC FOR PROJECT REPORTS

FML2ILV – Federated Machine Learning

ABSTRACT

This document contains the grading criteria for evaluating project reports in the FML2ILV Federated Machine Learning course. Reports are assessed by the instructor using the rubric below (total: 21 points; plagiarism may trigger a fail).

1. GRADING CRITERIA

The rubric follows the FL design workflow taught in the course: (i) formulate the application as GTVMin (Problem Formulation), (ii) characterize the solution via a fixed-point equation (Methodology), (iii) iterate the fixed-point operator to obtain an FL algorithm (Methodology), (iv) evaluate the algorithm empirically (Numerical Experiments).

1.1. Title and Introduction (2p)

1. Is the title informative and does the introduction outline the report?

- 1p – Title clearly describes the project’s focus *and* the introduction contains a concise outline of the report structure.
- 0p – Title is vague/misleading or outline is missing.

2. Does the introduction present a clear motivation and background?

- 1p – Introduction motivates the FL application in a real-world scenario and discusses (lack of) existing work.
- 0p – Motivation or background is missing or superficial.

1.2. Problem Formulation (5p)

Note: The raw data source (FMI Open Data) is fixed for all projects. The definition of local data points (features and labels), local loss functions, local models, and the edges of the FL network are design choices that each student must describe and justify.

3. Does the report clearly define the nodes, edges, and edge weights of the FL network and justify these choices?

- 2p – Nodes (including justification of which stations were selected), edges, and weights are clearly explained and justified as deliberate design choices.
- 1p – Some parts of the FL network are described but justification of the station subset or edge construction is missing.
- 0p – Description of FL network is insufficient.

4. Are features and labels defined and justified?

- 1p – Features and labels are clearly defined and their choice is justified (e.g. why these measurements, what prediction target, which time window); it is clear how local data points are derived from the raw FMI data.
- 0p – Features or labels are unclear or their choice is not justified.

5. Are the local models properly described and justified?

- 1p – Local models are specified (parameter space, hypothesis class) and the choice is justified.
- 0p – The description of models is unclear or the choice is not justified.

6. Are the local loss functions properly described and justified?

- 1p – The loss function is specified and justified (e.g. why MSE vs. MAE) as a design decision.
- 0p – The loss function is unclear or the choice is not justified.

1.3. Methodology (5p)

7. Does the report derive the fixed-point equation from the GTVMin formulation?

- 1p – The report states the fixed-point equation $\mathbf{w}^{(i)} = \mathcal{F}^{(i)}(\mathbf{w}^{(1)}, \dots, \mathbf{w}^{(n)})$ (or an equivalent form) and explains the block-wise minimization operator $\mathcal{F}^{(i)}$ at each node.
- 0p – The fixed-point equation is missing or incorrect.

8. Does the report present the FL algorithm as iteration of the fixed-point operator?

- 2p – The algorithm is presented as the iteration $\mathbf{w}^{(i,t+1)} = \mathcal{F}^{(i)}(\mathbf{w}^{(1,t)}, \dots, \mathbf{w}^{(n,t)})$, including the synchronous message-passing structure between neighbours.
- 1p – The algorithm is stated but not clearly linked to the fixed-point operator, or message passing is omitted.
- 0p – The algorithm description is missing or unclear.

9. Does the report explain how $\mathcal{F}^{(i)}$ is implemented via a scikit-learn `.fit()` call?

- 1p – The report names a specific scikit-learn class (e.g., Ridge for ℓ_2 -regularized linear models, LinearRegression, SGDRegressor) and explains how the local data, regularization parameter α , and messages from neighbours are mapped to its constructor and `.fit()` arguments.
- 0p – The implementation of $\mathcal{F}^{(i)}$ via `.fit()` is unclear or missing.

10. Is the GTVMin / fixed-point formulation correct?

- 1p – Dimensions, indices, and the relation between local losses, GTV penalty, regularization parameter α , and fixed-point operator $\mathcal{F}^{(i)}$ are consistent.
- 0p – The formulation contains errors (e.g., dimension mismatch, ill-defined loss, undefined edge weights, inconsistent operator).

1.4. Numerical Experiments (5p)

11. Does the report explain model validation and selection?

- 1p – The train/validation/test split is explicitly described and the splitting strategy suits the data structure (e.g., chronological for time series, spatial for geographic data).
- 0p – Validation is missing, lacks detail, or the strategy is inappropriate.

12. Are the training, validation, and test losses reported and discussed?

- 1p – Loss values are presented for training, validation, and test sets with analysis.
- 0p – Loss reporting is unclear, incomplete, or lacks discussion.

13. Does the report compare at least two distinct FL systems against a meaningful baseline?

- 2p – Two or more FL systems differing in a structural design choice (e.g., different graph constructions, local models, or loss functions — not just hyper-parameters such as α) are compared with quantitative results, *and* compared against a baseline such as local-only training ($\alpha = 0$) or a single shared model ($\alpha \rightarrow \infty$).
- 1p – Two systems are compared but no baseline is included, or systems differ only in hyper-parameters.
- 0p – Only a single FL system is evaluated.

14. Are figures and tables clear, self-contained, and informative?

- 1p – Figures have labelled axes, legends, informative captions; tables are well-structured; visualizations effectively convey the intended message.
- 0p – Figures or tables are missing, unlabelled, or difficult to interpret.

1.5. Reproducibility (1p)

15. Does the report list the implementation details required for reproduction?

Specifically:

- (a) exact FMI station IDs (or names) and time range used,
- (b) regularization parameter α , learning rate η , number of iterations / convergence threshold,
- (c) initialization of local model parameters and any random seed.

- 1p – All three items are clearly stated.
- 0p – One or more items are missing.

Practical test: pasting the relevant sections into an LLM (e.g. Claude Code) should produce a working implementation of the FL algorithm without guessing any missing details.

1.6. Conclusion (1p)

16. Does the conclusion summarise findings and discuss limitations?

- 1p – The conclusion concisely summarizes the main quantitative results *and* identifies concrete limitations with specific suggestions for improvement.
- 0p – The summary or discussion of limitations is missing or generic.

1.7. Formatting (1p)

17. Does the report comply with the formatting requirements?

Check all of the following:

- (a) The report does not exceed 5 pages (5th page contains only references).
 - (b) Margins, font sizes, and section titles are unchanged from the template.
- 1p – All formatting requirements are met.
 - 0p – One or more requirements are violated.

1.8. Plagiarism / Academic Integrity (1p)

18. Are all sources properly cited?

- 1p – All sources are correctly cited; no signs of plagiarism.
- 0p – Some citations are missing or incomplete.
- **FAIL** – Suspected plagiarism (verbatim copying, undisclosed AI generation of substantial passages, or unattributed reuse of others' work) triggers an academic integrity review per the IMC Krems Study and Examination Regulation. The project is then graded as failed independently of the rubric score above.

Score Summary

Section	Points
Title and Introduction	2
Problem Formulation	5
Methodology (GTVMin / fixed-point)	5
Numerical Experiments	5
Reproducibility	1
Conclusion	1
Formatting	1
Citations	1
Total	21
Plagiarism trigger	FAIL